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B.Tech. Degree V Semester Special Supplementary Examination April 2023

CE/CS/EC/EE/IT/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve algebraic and transcendental equations by numerical methods.

CO2: Solve numerical differentiation and integration problems.

CO3: Compute the mean and variance of a probability distribution including the binomial distribution.

CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

- | | (8 × 3 = 24) | Marks | BL | CO | PO |
|--|--------------|-------|----|----|----|
| 1. (a) Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using trapezoidal rule by taking $h = 0.2$.
Compare this with the exact value. | 3 | | L2 | 2 | 2 |
| (b) If X follows Poisson distribution such that $P(X=2) = \frac{2}{3}P(X=1)$,
then find $P(X=3)$ and $P(X \geq 3)$. | 3 | | L2 | 3 | 3 |
| (c) Using Taylor series method find $y(0.2)$ for the initial value problem
$\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$. | 3 | | L3 | 2 | 2 |
| (d) If a random sample of size n with sample mean $\bar{X}$ is taken from an
infinite population having the mean μ and variance σ^2 then show that
$E(\bar{X}) = \mu$ and $V(\bar{X}) = \frac{\sigma^2}{n}$. | 3 | | L1 | 4 | 1 |
| (e) Using Newton-Raphson's method, find the double root of the equation
$x^3 - 5x^2 + 8x - 4 = 0$ which is near 1.8. | 3 | | L2 | 1 | 2 |
| (f) Given that the probability density function of a random variable X is
$f(x) = \begin{cases} (k+2)x^3 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$. Then find the value of k and $V(X)$. | 3 | | L1 | 3 | 2 |
| (g) In 64 randomly selected hours of production, the mean and standard
deviation of the number of acceptable pieces produced by an
automatic stamping machine are $\bar{x} = 1038$ and $s = 146$. At the 0.05
level of significance, does this enable us to reject the null hypothesis
$H_0: \mu = 1000$ against the alternative hypothesis $H_1: \mu > 1000$. | 3 | | L2 | 4 | 3 |
| (h) Evaluate $f(4)$ from the following data:
$f(0) = 1, f(1) = 2, f(2) = 1, f(3) = 10$. | 3 | | L3 | 1 | 2 |

(P.T.O.)

PART B

(4 × 12 = 48) Marks BL CO PO

- II. (a) Find the root of the equation $xe^x = \cos x$ using Regula-falsi method correct to three decimal places. 7 L2 1 2

- (b) Using Lagrange's formula find $f(9)$ from the following data: 5 L3 1 2

5	7	11	13	17
150	392	1452	2366	5202

OR

- III. (a) Given 6 L2 1 2

θ°	0	5	10	15	20	25	30
$\tan \theta$	0	0.0875	0.1763	0.2679	0.3640	0.4663	0.5774

Using Stirling's formula, estimate the value of $\tan 16^\circ$.

- (b) Determine $f(x)$ as a polynomial in x for the following data using Newton's divided difference formula. 6 L3 1 2

x	-4	-1	0	2	5
$f(x)$	1245	33	5	9	1335

- IV. (a) Given that 6 L3 2 2

x	4.0	4.2	4.4	4.6	4.8	5.0	5.2
$\log x$	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

Evaluate $\int_4^{5.2} \log x dx$ by (i) Simpson's $\frac{1}{3}$ rule (ii) Simpson's $\frac{3}{8}$ rule.

- (b) Given $\frac{dy}{dx} = \log(x+y)$, $y(0) = 2$. Using modified Euler's method find the value of y at (i) $x = 1.0$ (ii) $x = 1.2$ by taking $h = 0.2$. 6 L2 2 2

OR

- V. (a) Using the Runge-Kutta method of fourth order solve for y at $x = 1.2$ and $x = 1.4$ for $\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}$, $y(1) = 0$. 6 L3 2 2

- (b) The table below reveals the velocity v of a body during the specified time t . Find its acceleration at $t = 1.1$. 6 L2 2 3

t	1.0	1.1	1.2	1.3	1.4
v	43.1	47.7	52.1	56.4	60.8

- VI. (a) The mean weight of 500 students in a certain college is 151 lb and standard deviation is 15 lb. Assuming the weights are normally distributed find how many students weigh:
(i) between 120 and 150 lb. (ii) more than 185 lb. 6 L3 3 3

- (b) Fit a second degree curve for the following data: 6 L3 3 2

x	1	2	3	4	5
y	3	9	13	21	31

OR

- VII. (a) Derive the mean and variance of binomial distribution. 6 L1 3 1
(b) In a certain factory producing razor blades there is 0.2% chance for any blade to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing: (i) no defective (ii) one defective (iii) more than two defectives in a consignment of 10,000 packets. 6 L2 3 2

BTS-V(SS)-04-23-1752

VIII. (a) In a random sample of 100 tube lights produced by company A, the mean lifetime of tube light is 1190 hours with standard deviation of 90 hours. Also in a random sample of 75 tube lights from a company B, the mean lifetime is 1230 hours with standard deviation of 120 hours. Is there a difference between the mean lifetime of the two brands of tube light?

6 L3 4 3

(b) Two independent random samples of size 12 are taken from two gold manufacturing companies. Determine whether there is less variability in the production done by Company I than in that done by company II, if the samples from the two companies' work yield $s_1 = 0.035$ and $s_2 = 0.062$.

6 L4 4 4

OR

IX. (a) The standard deviation of the tube's life for a particular brand of ultraviolet tubes is 500 hours and the operating life of the tubes is normally distributed. The manufacturer claims that average tube's life is at least 9000 hours. Test the claim at 5% level of significance, given that a random sample of 40 tubes gave a mean life of 8800 hours.

6 L3 4 4

(b) The standard deviation of a sample of size 16 from a normal population was found to be 7. Examine whether the hypothesis that the standard deviation is 7.6 is acceptable at 5% level of significance.

6 L4 4 3

Bloom's Taxonomy Levels

L1 = 10%, L2 = 35.83%, L3 = 44.17%, L4 = 10%.

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B.Tech. Degree V Semester Special Supplementary Examination September 2022

CE/CS/EC/EE/IT/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS
(2019 Scheme)

Maximum Marks: 60

Time: 3 Hours

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve algebraic and transcendental equations by numerical methods.

CO2: Solve numerical differentiation and integration problems.

CO3: Compute the mean and variance of a probability distribution including the binomial distribution.

CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Prove that (i) $\mu^2 = 1 + \frac{\delta^2}{4}$ (ii) $\mu\delta = \frac{1}{2}(\Delta + \nabla)$.	3	L1, L3	1	1
(b) Find the third divided difference of $f(x)$ with arguments 2, 4, 9, 10 where $f(x) = x^3 - 2x$.	3	L3	1	1
(c) Evaluate $\int_0^1 \frac{dx}{(1+x^2)}$ using Simpson's $\frac{1}{3}$ rule taking $h = \frac{1}{4}$.	3	L1, L3	1	3
(d) Use Euler's method to find $y(0.4)$ given $y' = xy$, $y(0) = 1$ (Take $h = 0.2$)	3	L3	2	2
(e) A random variable x has the following probability function.	3	L1, L3	3	1
$\begin{array}{cccccc} x: & -2 & -1 & 0 & 1 & 2 & 3 \\ p(x): & 0.1 & k & 0.2 & 2k & 0.3 & k \end{array}$				
Find the value of k and calculate mean and variance.				
(f) Find the probability that no defective fuse will be found in a box of 200 fuses if experience shows that 2% such fuses are defective.	3	L3, L4	3	3
(g) Define (i) parameter and statistic (ii) critical region (iii) test statistic.	3	L1	4	1
(h) The average score in mathematics of a sample of 100 students was 51 with a standard deviation of 6 points. Could this have been a random sample from a population with average score 50?	3	L4	4	4

PART B

(4 × 12 = 48)

II. (a) Find a real root of $x^3 + x - 1 = 0$ near $x = 1$ correct to 3 decimal places by Regula-Falsi method.	6	L3	1	2
(b) Solve the system of equation by Gauss Seidel method.	6	L3	1	2
$\begin{array}{l} 27x + 6y - z = 85 \\ 6x + 15y + 2z = 72 \\ x + y + 54z = 110 \end{array}$				

OR

(P.T.O.)

BTS-V(SS)-09-22-1059

Marks	BL	CO	PO
6	L3	1	2

III. (a) Using Lagrange's formula find $f(6)$ from the following table.

$x:$	2	5	7	10	12
$f(x):$	18	180	448	1210	2028

(b) From the following table of half-yearly premium for policies maturing at different ages, find the premium for policies maturing at age 46 and 63.

Age $x:$	45	50	55	60	65
Premium $y:$	1150	960	830	740	680

IV. (a) Using modified Euler method, find $y(0.2)$, $y(0.4)$, given $y' = y - x^2$ $y(0) = 1$.

(b) The table below gives the results of an observation: θ is the observed temperature in degree centigrade of a vessel of cooling water; t is the time in minutes from the beginning of observation.

$t:$	1	3	5	7	9
$\theta:$	85.3	74.5	67.0	60.5	54.3

Find the approximate rate of cooling at $t = 3$.

OR

V. (a) Apply Runge-Kutta method to obtain $y(0.2)$, given $y' = \frac{y-x}{y+x}$ when $y(0) = 1$.

(b) A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is given in the table below. Using

Simpson's $\frac{1}{3}$ rule, find the velocity of the rocket at $t = 80$ seconds.

$t(\text{sec}):$	0	10	20	30	40	50	60	70	80
$A(\text{cm/sec}^2):$	30.0	31.63	33.34	35.47	37.75	40.33	43.25	46.69	50.67

VI. (a) Derive the mean and variance of Binomial Distribution.

(b) A screw manufacturing company is known to produce 5% defectives in a random sample of 15 screws. What is the probability that there are (i) exactly 3 defectives (ii) not more than 3 defectives

OR

VII. (a) Fit a parabola, by the method of least squares, to the following data.

$x:$	1	2	3	4	5
$y:$	5	12	26	60	97

(b) A sample of 100 dry battery cells tested to find the length of life produces the following results: $\bar{x} = 12$ hrs, $\sigma = 3$ hrs. Assuming the data to be normally distributed, what percentage of battery cells are expected to have life (i) more than 15 hrs (ii) less than 6 hrs (iii) between 10 and 14 hrs

VIII. (a) A manufacturer of tyres guarantees that the average lifetime of its tyres is more than 28000 miles. If 40 tyres of this company tested, yields a mean lifetime of 27463 miles with standard deviation of 1348 miles, can the guarantee be accepted at 0.01 level of significance?

(b) A sample of 900 numbers is found to have a mean 3.4 and standard deviation 2.61. Is this sample taken from a large population of mean 3.25?

OR

(Continued)

BTS-V(SS)-09-22-1059

		Marks	BL	CO	PO
IX.	(a)	6	L4	4	3
	(b)	6	L4	4	4

The standard deviation of a sample of 20 observations from a normal population was found to be 5. Examine whether the sample was taken from a population with standard deviation 5.3.

In a random sample of 100 tube lights produced by a company *A*, the mean lifetime of tube light is 1190 hrs with standard deviation of 90 hrs. Also in a random sample of 75 tube lights from company *B* the mean lifetime is 1230 hrs with standard deviation 120 hrs. Is there a difference between the mean lifetime of the two brands of the tube lights?

B.Tech. Degree V Semester Regular Examination November 2021**CS/CE/ EC/EE/IT/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS**
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer ALL questions)

(8 × 3 = 24)

- I. (a) Find a positive value of $(17)^{1/3}$ correct to three places of decimal using Newton-Raphson method. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ 2.571
- (b) Prove that,

$$(i) \mu^2 = 1 + \frac{\delta^2}{4}$$

$$(ii) 1 + \mu^2 \delta^2 = \left(1 + \frac{\delta^2}{2}\right)^2$$

- (c) Prove that,

$$h f^1(a) = \Delta f(a) - \frac{1}{2} \Delta^2 f(a) + \frac{1}{3} \Delta^3 f(a) \dots$$

- (d) $\frac{dy}{dx} = 1 + xy$, $y(0) = 1$ using Taylor series method. Compute $y(0.1)$ correct to 4 decimal places. 1.10534

- (e) An urn contains four white and three red balls. Three balls are drawn with replacement. Find the mean and variance of number of red balls drawn. $\frac{9}{7}, \frac{36}{49}$
- (f) By method of least squares find the best fitting straight line to the following data.

x:	0	1	2	3	4
y:	1	3	6	10	15

- (g) Define the following terms.

- (i) Parameter and statistic
- (ii) Type I and Type II errors
- (iii) Standard error. S.D of sampling distri of a statistic is known as S.E
- (h) Assuming the mean height of soldiers to be 68.22 inches with a variance of 10.8 inches, find how many soldiers in a regiment of 1000 would you expect to be over 6 feet tall

(P.T.O)

PART B

(4 × 12 = 48)

- II. (a) Find the solution of the following system of equations using Gauss-Seidal method.

$$83x + 11y - 4z = 95$$

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71$$

- (b) The population of a town is as follows:

Year	1961	1971	1981	1991	2001
Population (in thousands):	24	29	36	46	51

Estimate the increase in population during the period 1966 to 1996.

OR

- III. (a) Use Newton's dividend difference to find $f(7)$ and $f(10)$ if $f(3) = 24$, $f(5) = 120$, $f(8) = 504$, $f(9) = 720$, $f(12) = 1716$

- (b) Find a real root of $x^3 - 9x + 1 = 0$ by Regula-Falsi method.

- IV. (a) A slider machine moves along a fixed straight rod. It's distance x cm along the rod is given below for various values of time t sec. Find the velocity and acceleration when $t = 0.3$ sec.

t :	0.2	0.3	0.4	0.5	0.6
x :	32.87	33.64	33.95	33.81	33.24

- (b) Compute $\int_0^6 \frac{dx}{1+x^2}$ using

(i) Trapezoidal rule

(ii) Simpson's $\frac{1}{3}$ rule

OR

- V. (a) Use modified Euler's method to approximate y when $x = 0.1$ given

$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1 \text{ by taking } h = 0.1$$

- (b) By Runge-Kutta method of fourth order calculate the value of y for

$$x = 0.2 \text{ correct to 3 decimal places when } \frac{dy}{dx} = x - 2y, \quad y(0) = 0$$

taking $h = 0.1$

(Continued to 3)

BTS-V(R)-11-21-0937

- VI. (a) A Screw manufacturing company is known to produce 5% defectives in a random sample of 15 screws. What is the probability that there are
- Exactly 3 defectives
 - Not more than three defectives
- (b) Find mean and variance of Poisson distribution.

OR

- VII. (a) In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution.
- (b) Fit an equation of the form $y = ab^x$ to the following data:

x:	2	3	4	5	6
y:	144	173	207	249	299

- VIII. (a) The mean height of college students in a city are normally distributed with SD 6 cm. A sample of 100 students has mean height 158 cm. Test the hypothesis that mean height of students in the city is 160 cm.
- (b) Given the following data of two distributions

	Size of sample	Mean	SD
A	80	100	12
B	70	95	10

Test whether difference between the sample means is significant.

OR

- IX. (a) The SD of a sample of 20 observations from a normal population was found to be 5. Examine whether the sample was taken from a population with SD 5.3.
- (b) Two independent of size $n_1 = 10, n_2 = 7$ when observed have sample variance $s_1^2 = 16, s_2^2 = 3$ using $\alpha = 0.01$ Test $H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$

$F = 5.08, F_{\alpha} = 7.98$ Acc
